

through the process of sampling. How should a decision maker collect sample data, compute sample statistics, and use this information to ascertain the correctness of the hypothesized population parameter? Also, if Liberty wants to check whether the 18% market share attributed to the organized sector is correct, it can take a sample of consumers, compute sample statistic, and use this information to ascertain the correctness of the hypothesized population parameter. This chapter discusses the concept of hypothesis testing, two-tailed and one-tailed tests of hypothesis testing, concept of Type I and Type II errors in hypothesis testing, concept of hypothesis testing for a single population using z statistic, the concept of p -value approach and critical value approach for hypothesis testing, the concept of hypothesis testing for a single population using t statistic, and the procedure of hypothesis testing for population proportion.

10.1 INTRODUCTION

In Chapter 9, we have discussed how a sample can be used to develop point and interval estimates for assessing the population parameter. As discussed earlier, statistical inference is based on estimation and hypothesis testing. In this chapter, we will continue the discussion of statistical inference in terms of hypothesis testing. Hypothesis testing is the soul of inferential statistics and is a very important tool, which can be used by a business researcher to arrive at a meaningful conclusion. The main objective of this chapter is to discuss how hypothesis testing can be conducted around population parameters.

10.2 INTRODUCTION TO HYPOTHESIS TESTING

In diverse fields such as marketing, personnel management, financial management, etc. decision makers need right answers to certain questions in order to take optimum decisions. For example, a marketing manager might be interested in assessing the customer loyalty for a particular product; a personnel manager might be interested in knowing the job satisfaction level of employees; a financial manager might be interested in understanding the financial aspect of the company's retirement scheme, etc. In every case the concerned manager has to make decisions on the basis of the available information and in most cases, information is obtained through sampling. The sample statistic is computed through sampling and it is used to make an inference about the population parameters. These are just examples. In real life, managers encounter many situations where they need to find solutions to problems. As discussed earlier, a complete census is neither practical nor statistically recommended (due to non-sampling errors).

In order to find out the answers to these questions, a decision maker needs to collect sample data, compute the sample statistic and use this information to ascertain the correctness of the hypothesized population parameter. For this purpose, a researcher develops a "hypothesis" which can be studied and explored. For example, suppose the Vice President (HR) of a company wants to know the effectiveness of a training programme which the company has organized for all its 70,000 employees based at 130 different locations in the country. Contacting all these employees with an effectiveness measurement questionnaire is not feasible. So the Vice President (HR) takes a sample of size 629 from all the different locations in the country. The result that is obtained would not be the result from the entire population but only from the sample. The Vice President (HR) will then set an assumption that "training has not enhanced efficiency" and will accept or reject this assumption through a well-defined statistical procedure known as **hypothesis testing**. A statistical hypothesis is an assumption about an unknown population parameter. Hypothesis testing starts with an assumption termed as "hypothesis" that a researcher makes about a population parameter. We cannot accept or reject the hypothesis on the basis of intuition or on the basis of general information. **Hypothesis testing** is a well-defined procedure which helps us to decide objectively whether to accept or reject the hypothesis based on the information available from the sample.

Now we need to understand the rationale of hypothesis testing. Drawing a random sample from the population is based on the assumption that the sample will resemble the population. Based on this philosophy, the known sample statistic is used for estimating the unknown population parameter. When a researcher sets a hypothesis or assumption, he assumes that the sample statistic will be close to the hypothesized population parameter. This is possible in cases where the hypothesized population parameter is correct and the sample statistic is a good estimate of the population parameter. In real life, we cannot expect the sample statistic to always be a good estimate of the population parameter. Differences are likely to occur due to sampling and non-sampling errors or due to chance. A large

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difference between the sample statistic and the hypothesized population parameter raises questions on the accuracy of the sampling technique. In statistical analysis, we use the concept of probability to specify a probability level at which a researcher concludes that the observed difference between the sample statistic and the population parameter is not due to chance.

10.3 HYPOTHESIS TESTING PROCEDURE

We have already discussed that hypothesis testing is a well-defined procedure. A sample is selected for estimating the population parameter. Sample statistic is computed from this sample and is used to estimate the population parameter. A systematic procedure needs to be adopted for hypothesis testing. The seven steps in hypothesis testing are shown in Figure 10.1.

Step 1: Set null and alternative hypotheses

The null hypothesis, generally referred to as H_0 (H sub-zero), is the hypothesis which is tested for possible rejection under the assumption that it is true. Theoretically, a null hypothesis is set as no difference or status quo and considered true, until and unless it is proved wrong by the collected sample data. The null hypothesis is always expressed in the form of an equation, which makes a claim regarding the specific value of the population. Symbolically, a null hypothesis is represented as

$$H_0: \mu = \mu_0$$

where μ is the population mean and μ_0 is the hypothesized value of the population mean. For example, to test whether a population mean is equal to 150, a null hypothesis can be set as “population mean is equal to 150.” Symbolically,

$$H_0: \mu = 150$$

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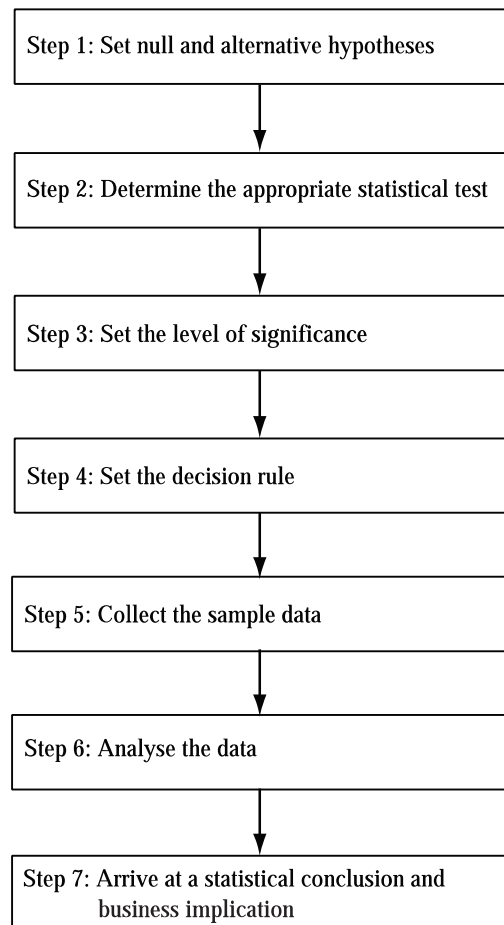


FIGURE 10.1
Seven steps of hypothesis testing

The alternative hypothesis, generally referred by H_1 (H sub-one), is a logical opposite of the null hypothesis. In other words, when null hypothesis is found to be true, the alternative hypothesis must be false or when the null hypothesis is found to be false, the alternative hypothesis must be true.

The **alternative hypothesis**, generally referred by H_1 (H sub-one), is the logical opposite of the null hypothesis. In other words, when null hypothesis is found to be true, the alternative hypothesis must be false or when null hypothesis is found to be false, the alternative hypothesis must be true. Symbolically, alternative hypothesis is represented as:

$$H_1: \mu \neq \mu_0$$

Consequently,

$$H_1: \mu < \mu_0$$

$$H_1: \mu > \mu_0$$

For the above example, the alternative hypothesis can be set as “population mean is not equal to 150.” Symbolically,

$$H_1: \mu \neq 150$$

This results in two more alternative hypotheses, $H_1: \mu < 150$, which indicates that the population mean is less than 150 and $H_1: \mu > 150$; which indicates that the population mean is greater than 150.

Step 2: Determine the appropriate statistical test

After setting the hypothesis, the researcher has to decide on an appropriate statistical test that will be used for statistical analysis. Type, number, and the level of data may provide a platform for deciding the statistical test. Apart from these, the statistics used in the study (mean, proportion, variance, etc.) must also be considered when a researcher decides on appropriate statistical test, which can be applied for hypothesis testing in order to obtain the best results.

Step 3: Set the level of significance

The **level of significance** generally denoted by α is the probability, which is attached to a null hypothesis, which may be rejected even when it is true. The level of significance is also known as the size of the rejection region or the size of the critical region. It is very important to note that the level of significance must be determined before we draw samples, so that the obtained result is free from the choice bias of a decision maker. The levels of significance which are generally applied by researchers are: 0.01; 0.05; 0.10. The concept of “level of significance” is discussed in detail later in this chapter.

Step 4: Set the decision rule

The next step for the researcher is to establish a **critical region**, which is the area under the normal curve, divided into two mutually exclusive regions (shown in Figure 10.2). These regions are termed as acceptance region (when the null hypothesis is accepted) and the rejection region or critical region (when the null hypothesis is rejected).

If the computed value of the test statistic falls in the acceptance region, the null hypothesis is accepted, otherwise it is rejected. For making a decision regarding the acceptance or rejection of the null

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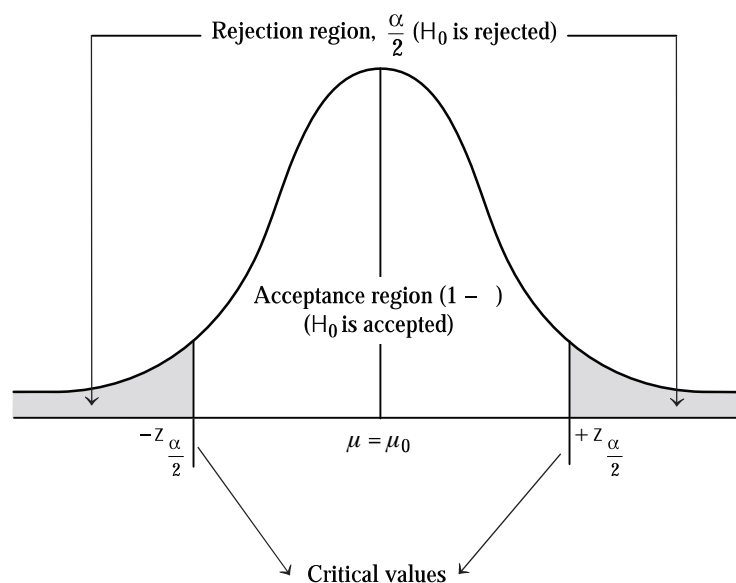


FIGURE 10.2
Acceptance and rejection regions of null hypothesis (two-tailed test)

hypothesis, a researcher has to determine the critical value which separates the rejection region from the acceptance region. The determination of critical value depends on the size of the rejection region, which is directly related to the risk involved in decision making. In other words, we can say that the size of the rejection region is directly related to the level of precision which a decision maker wants to maintain while estimating the population parameter.

Step 5: Collect the sample data

In this stage of sampling, data are collected and the appropriate sample statistics are computed. The first four steps should be completed before collecting the data for the study. It is not advisable to collect the data first and then decide on the stages of hypothesis testing. We have already discussed the process of sampling in Chapter 8.

Step 6: Analyse the data

In this step, the researcher has to compute the test statistic. This involves selection of an appropriate probability distribution for a particular test. For example, when the sample is small ($n < 30$), the use of the normal probability distribution (z) is not an accurate choice. The t distribution needs to be used in this case. Some of the commonly used testing procedures are z , t , F , and χ^2 . The selection of a suitable test statistic will be discussed in detail in the succeeding chapters.

Step 7: Arrive at a statistical conclusion and business implication

In this step, the researchers draw a statistical conclusion. A statistical conclusion is a decision to accept or reject a null hypothesis. This depends on whether the computed test statistic falls in the acceptance region or the rejection region. If we test a hypothesis at 5% level of significance and the observed set of results have a probability of less than 5%, we consider that the difference between the sample statistic and the hypothesized population parameter as significant. In this situation, a researcher decides to reject the null hypothesis and accept the alternative hypothesis. On the other hand, if the observed set of results have the probability of more than 5%, we consider the difference between the sample statistic and hypothesized population parameter as not significant. In this situation, a researcher decides to accept the null hypothesis and the alternative hypothesis is automatically rejected.

Statisticians present the information obtained using hypothesis-testing procedure to the decision makers. Decisions are made on the basis of this information. Ultimately, a decision maker decides that a statistically significant result is a substantive result and needs to be implemented for meeting the organization's goals.

If the computed value of the test statistic falls under the acceptance region, the null hypothesis is accepted, otherwise it is rejected. For making a decision regarding the acceptance or rejection of the null hypothesis, a researcher has to determine the critical value which separates the rejection region from the acceptance region.

10.4 TWO-TAILED AND ONE-TAILED TESTS OF HYPOTHESIS

There are two types of tests of hypothesis. They are two-tailed tests and one-tailed tests of hypothesis. Hypothesis formulation provides a base for test selection. This will be discussed in detail in the succeeding chapters.

10.4.1 Two-Tailed Test of Hypothesis

Let us consider a null and alternative hypotheses as below:

$$H_0: \mu = \mu_0$$

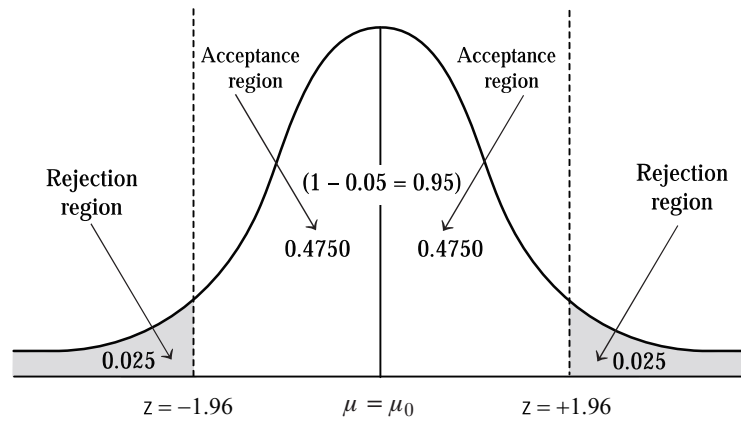
$$H_1: \mu \neq \mu_0$$

A two-tailed test contains the rejection region on both the tails of the sampling distribution of a test statistic. This means a researcher will reject the null hypothesis if the computed sample statistic is significantly higher than or lower than the hypothesized population parameter (considering both the tails, right as well as left). If the level of significance is α , then the rejection region will be on both the tails of the normal curve, consisting of $\frac{\alpha}{2}$ area on both the tails of the normal curve. For example, for testing a hypothesis at 5% level of significance, the size of acceptance region on each side of the mean will be 0.475 ($0.475 + 0.475 = 0.95$), and the size of rejection region on both the tails will be 0.025 ($0.025 + 0.025 = 0.05$). This is also shown in Figure 10.3.

When we consult the standard normal table, we find that the area 0.475 corresponds to 1.96 standard error on each side of μ_0 , which is equal to the size of the acceptance region (0.95% area). If a sample statistic falls outside this area, the null hypothesis is rejected. This is equal to the size of the rejection region (0.05% area). Similarly, for testing a hypothesis at 1% level of significance, the size

Two-tailed tests contain the rejection region on both the tails of the sampling distribution of a test statistic. This means a researcher will reject the null hypothesis if the computed sample statistic is significantly higher than or lower than the hypothesized population parameter (considering both the tails, right as well as left).

FIGURE 10.3
Acceptance and rejection regions ($\alpha = 0.05$)



of the acceptance region on each side of the mean will be 0.495 ($0.495 + 0.495 = 0.99$) and the size of rejection region on both the tails will be 0.005 ($0.005 + 0.005 = 0.01$). From the standard normal table, we find that the area 0.495 corresponds to 2.575 standard error on each side of μ_0 , and this is equal to the size of the acceptance region (0.99% area). This is shown in Figure 10.4.

When we compare Figures 10.3 and 10.4, a very important result emerges. When we decrease the size of the rejection region (Figure 10.4), the probability of accepting a null hypothesis increases.

10.4.2 One-Tailed Test of Hypothesis

Let us consider a null and alternative hypotheses as below:

$$H_0: \mu = \mu_0 \quad \text{and} \quad H_1: \mu < \mu_0 \quad (\text{Left-tailed test})$$

$$H_0: \mu = \mu_0 \quad \text{and} \quad H_1: \mu > \mu_0 \quad (\text{Right-tailed test})$$

Unlike the two-tailed test, the one-tailed test contains the rejection region on one tail of the sampling distribution of a test statistic. In case of a left-tailed test, a researcher rejects the null hypothesis if the computed sample statistic is significantly lower than the hypothesized population parameter (considering the left side of the curve in Figure 10.5). In the case of a right-tailed test, a researcher rejects the null hypothesis if the computed sample statistic is significantly higher than the hypothesized population parameter (considering the right side of the curve in Figure 10.6). Thus, in a one-tailed test the entire rejection region corresponding to the level of significance (α) is located only in one tail of the sampling distribution of the statistic.

Let us discuss the example which we had taken up in the section discussing “hypothesis testing procedure” with population mean equal to 150. As discussed in this example, the null hypothesis is

$$H_0: \mu = 150$$

A two-tailed alternative hypothesis (population mean is not equal to 150) can be exhibited as below

$$H_1: \mu \neq 150 \quad (\text{Two-tailed test})$$

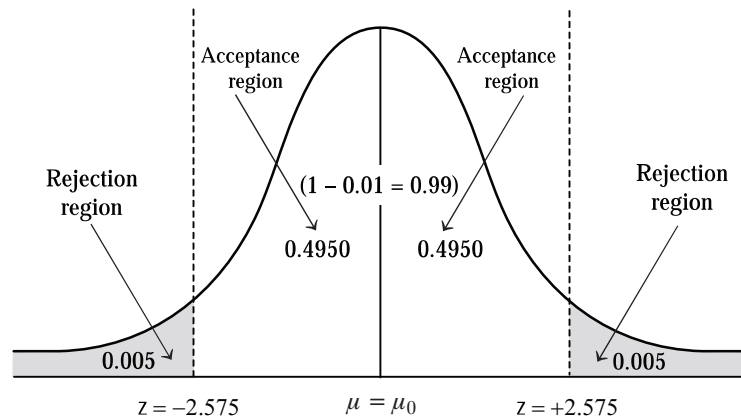


FIGURE 10.4
Acceptance and rejection regions ($\alpha = 0.01$)

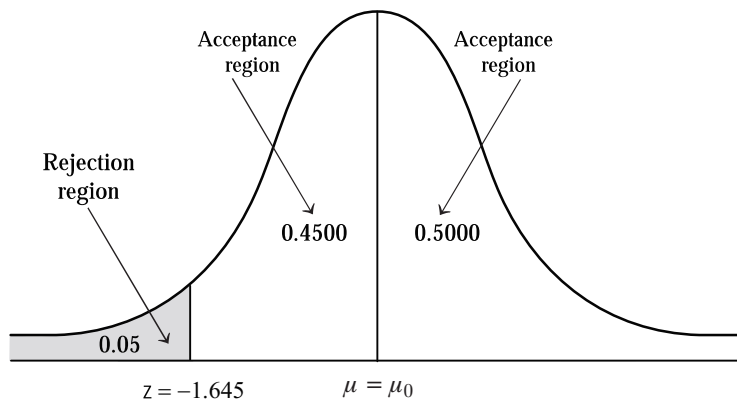


FIGURE 10.5
Acceptance and rejection
regions for one-tailed (left)
test ($\alpha = 0.05$)

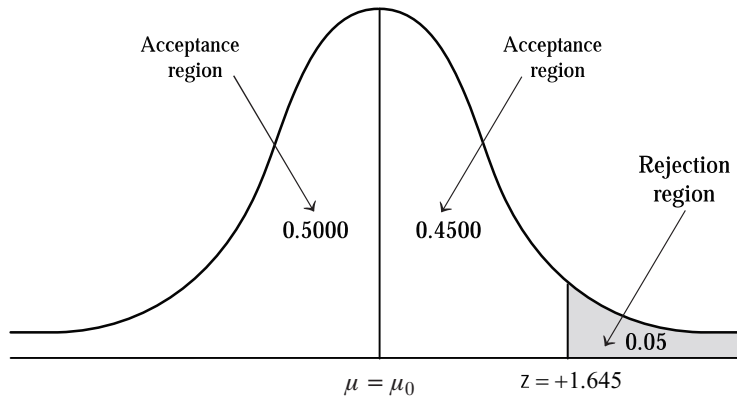


FIGURE 10.6
Acceptance and rejection
regions for one-tailed (right)
test ($\alpha = 0.05$)

A one-tailed (left) alternative hypothesis (population mean is less than 150) can be exhibited as below

$$H_1 : \mu < 150 \text{ (Left-tailed test)}$$

A one-tailed (right) alternative hypothesis (population mean is more than 150) can be exhibited as below

$$H_0 : \mu > 150 \text{ (Right-tailed test)}$$

Table 10.2 shows a summary of certain values at various significance levels for test statistic z .

10.5 TYPE I AND TYPE II ERRORS

While testing hypotheses, null hypothesis should be accepted when it is true and null hypothesis should be rejected when it is false. In a real-life situation, the correct decision is not always possible. We know that the hypothesis-testing procedure uses a sample statistic (based on the sample) to arrive at a conclusion about the population parameter. In this process, the possibility of making incorrect decisions about the null hypothesis cannot be ignored. In fact, when a researcher tests statistical hypotheses, there can be four possible outcomes as follows:

TABLE 10.2
Values of z_α for the most commonly used confidence intervals

Confidence level ($1 - \alpha$) %	(α)	One-tailed region	Two-tailed region
90%	0.10	± 1.28	± 1.645
95%	0.05	± 1.645	± 1.96
99%	0.01	± 2.33	± 2.575

TABLE 10.3

Errors in hypothesis testing and power of the test

Statistical decision	State of nature	
	H_0 True	H_0 False
Accept H_0	Correct decision with confidence level $(1 - \alpha)$	Type II error, $P(\text{Type II error}) = \beta$
Reject H_0	Type I error, $P(\text{Type I error}) = \alpha$	Correct decision, Power of the test $= (1 - \beta)$

1. Rejecting a true null hypothesis (Type I error)
2. Accepting a false null hypothesis (Type II error)
3. Accepting a true null hypothesis (Correct decision)
4. Rejecting a false null hypothesis (Correct decision)

1. A Type I error is committed by rejecting a null hypothesis when it is true. For example, a quality control manager of a ballpoint pen manufacturing firm finds that 5% pens are defective. For testing the hypothesis, researchers take a random sample of 100 pens and test to find the defective pieces. It is possible that this sample contains the most extreme lots (more than 10% or less than 2%), leading to the rejection of the null hypothesis though the population mean is actually 5%. In this case, the researchers have committed Type I error. The possibility of committing Type I error is called (α) or level of significance. α is the area under the curve which is in the rejection region beyond the critical values. As discussed earlier, some of the most common values of α are 0.10, 0.05, and 0.01.

2. A Type II error is committed by accepting a null hypothesis when it is false. In the example related to ballpoint pens, suppose the population mean is 6% defective pieces even though the null hypothesis is 5% defectives. A sample of 100 pens yields 5.2% defectives which falls in the non-rejection region. The researcher does not reject the null hypothesis (he accepts a false null hypothesis). In this situation, a Type II error is committed. The probability of committing type II error is beta (β) . Symbolically,

α = Probability of committing Type I error

β = Probability of committing Type II error

We need to examine the relationship between (α) and (β) . Type I error is committed by rejecting a null hypothesis when it is true, and Type II error is committed by accepting a null hypothesis when it is false. A researcher cannot commit Type I and Type II errors at the same time on the same hypothesis test. Generally, α and β are inversely related to each other, that is, if α is reduced, β is increased and if β is reduced, α is increased. $(1 - \beta)$ is the power of the test and measures the probability of rejecting the false null hypothesis. Table 10.3 exhibits the relationship between the two types of errors, confidence level, and the power of a test.

10.6 HYPOTHESIS TESTING FOR A SINGLE POPULATION MEAN USING THE z STATISTIC

Hypothesis testing for large samples ($n \geq 30$) is based on the assumption that the population, from which the sample is drawn, has a normal distribution. As a result, the sampling distribution of mean $\bar{x}$ is also normally distributed. Even when the population is not normal, the sampling distribution of mean $\bar{x}$ for a large sample size is normally distributed, irrespective of the shape of the population (central limit theorem). For testing hypothesis about a single population mean, z formula can be used, if the sample size is large ($n \geq 30$) for any population; for small samples ($n < 30$), if x is normally distributed. As discussed earlier, z formula can be stated as below:

z Formula for a single population mean

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

where μ is the population mean, σ the population standard deviation, n the sample size, and $\bar{x}$ the sample mean.

A Type I error is committed by rejecting a null hypothesis when it is true. The possibility of committing Type I error is called (α) or the level of significance. α is the area under the curve which is in the rejection region beyond the critical values.

A Type II error is committed by accepting a null hypothesis, when it is false. The probability of committing Type II error is beta (β) .

A marketing research firm conducted a survey 10 years ago and found that the average household income of a particular geographic region is Rs 10,000. Mr Gupta, who has recently joined the firm as a vice president has expressed doubts about the accuracy of the data. For verifying the data, the firm has decided to take a random sample of 200 households that yield a sample mean (for household income) of Rs 11,000. Assume that the population standard deviation of the household income is Rs 1200. Verify Mr Gupta's doubts using the seven steps of hypothesis testing. Let $\alpha = 0.05$.

Example 10.1

Solution

In the previous section, we have already discussed the seven steps of testing hypothesis. The first step is to establish the null and alternative hypotheses.

Step 1: Set null and alternative hypotheses

In this particular example, the researcher is trying to verify whether there is any change in the average household income within 10 years. The null hypothesis is set as no difference or status quo, that is, the average household income has not changed. Symbolically, this is given as

$$H_0: \mu = 10,000$$

The alternative hypothesis is a logical opposite of the null hypothesis. Hence,

$$H_1: \mu \neq 10,000$$

Step 2: Determine the appropriate statistical test

At this stage, an appropriate statistical test must be determined. In this case, sample size is large (≥ 30) and the sample mean is used as a statistic, so the z formula can be used for hypothesis testing. As discussed, z formula for a single population mean is given as

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Step 3: Set the level of significance

The level of significance α is also known as the size of rejection region or the size of the critical region and is 0.05 in this case.

Step 4: Set the decision rule

In the light of the alternative hypothesis, we are clear that this is a case of a two-tailed test (mean household income can be less than 10,000 and can be more than 10,000) and the level of significance is 0.05. As shown in Figure 10.7, the acceptance region covers 95% of the area and the rejection region covers the remaining

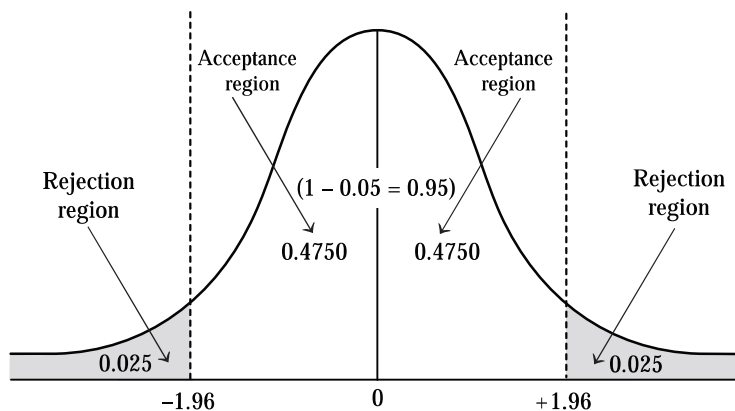


FIGURE 10.7
Acceptance and rejection region ($\alpha = 0.05$)

5% of the area at the two ends of the distribution. The critical z values can be obtained from the normal table as below:

$$z_{\frac{\alpha}{2}} = \pm 1.96$$

If the computed test statistic is between +1.96 and -1.96, the decision is to accept the null hypothesis and if the computed test statistic is outside ± 1.96 , the decision is to reject the null hypothesis (accept the alternative hypothesis).

Step 5: Collect the sample data

At this stage, a researcher collects the data. In this example, a sample of 200 respondents yields a sample mean of Rs 11,000.

Step 6: Analyse the data

The value of the sample statistic is calculated in this stage. From the example, $n = 200$, $\bar{x} = 11,000$, $\sigma = 1200$, and the hypothesized mean $\mu = 10,000$. z formula for a single population mean is as follows:

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

By substituting all the values we get: $z = \frac{11,000 - 10,000}{\frac{1200}{\sqrt{200}}} = 11.79$

Step 7: Arrive at a statistical conclusion and business implication

The calculated z value is 11.79, which is greater than +1.96. Hence, the statistical conclusion is to reject the null hypothesis and accept the alternative hypothesis. The calculated z value is also termed as the observed value. So, in this case the observed value of z is 11.79 and the critical value of z is +1.96.

On the basis of hypothesis testing, it can be concluded that the average household income of the particular geographic region is not Rs 10,000. Mr Gupta's doubts about this average household income was right. In the last 10 years, the average household income has increased. Since Rs 11,000 is only the sample mean, there is no guarantee that all the different samples taken from the population will produce an increase of Rs 1000 (confidence is 95%). However, broadly we can conclude that the average household income has increased and now policies of the company must be decided on the basis of this increased household income.

Note 1: In many real-life situations, the population standard deviation remains unknown. In this case, for a large sample size ($n \geq 30$), the population standard deviation (σ) can be replaced by a sample standard deviation (s) and the resulting z formula will be as under:

$$z = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Note 2: The z formula discussed above is based on the assumption that the sample is drawn from an infinite population. We have already discussed that in case of finite population, z formula must be modified by incorporating a finite correction factor. When the sample size is less than 5% of the population, the finite correction factor does not significantly increase the accuracy of the solution. In case of a finite population, z formula will be as below:

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}} \times \sqrt{\frac{N-n}{N-1}}}$$

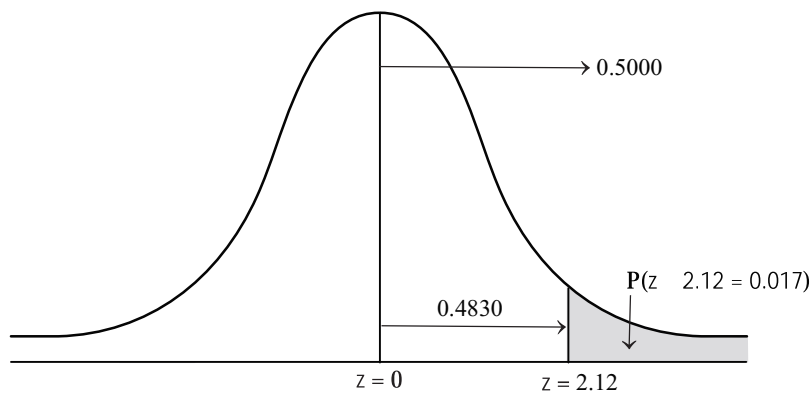


FIGURE 10.8
Probability of $p(z \geq 2.12)$

10.6.1 p -Value Approach for Hypothesis Testing

The p -value approach for hypothesis testing is used for large samples and is sometimes referred to as the observed level of significance. This approach is very advantageous especially after the introduction of many statistical software programs. The p value defines the smallest value of α for which the null hypothesis can be rejected. The decision rule for accepting or rejecting a null hypothesis based on the p value is as given below:

Reject the null hypothesis H_0 when the p value is $< \alpha$, otherwise, accept the null hypothesis H_0 .

For example, a researcher conducting a hypothesis test with rejection region on the right tail of the distribution obtains an observed test statistic value of 2.12. From the normal table, the corresponding probability area for z value 2.12 is 0.4830. So, the probability of obtaining a z value greater than or equal to 2.12 is $0.5000 - 0.4830 = 0.017$ (shown in Figure 10.8).

In this case, the p value is 0.017. For $\alpha = 0.05$ and $\alpha = 0.1$, this p value falls under the rejection region (at $\alpha = 0.05$, $0.017 < 0.05$ and $\alpha = 0.1$, $0.017 < 0.1$), so the null hypothesis will be rejected at 0.05 and 0.1 levels of significance. At $\alpha = 0.01$, the researcher cannot reject the null hypothesis for the value of α equal to 0.017 because $\alpha = 0.01 < 0.017$.

The procedure of calculating the p value for a two-tailed test is slightly different. In a two-tailed test, the rejection region falls in both the tails of the normal distribution. For example, at $\alpha = 0.05$, the rejection region is located in both the tail areas of the distribution in terms of 0.025% ($0.025 + 0.025 = 0.05$) area in both the tails of the distribution. The researcher compares this α value to the computed p value for accepting or rejecting the null hypothesis. For this comparison, instead of splitting the α value, we double the p value and then compare that p value with the α value. In the previous example, the researcher conducted a hypothesis test with rejection region on both the tails of the distribution (two-tailed test) and obtained the observed test statistic as 2.12 and corresponding p value as 0.017. So, instead of splitting α value in two parts, the researcher should double the p value, that is, ($0.017 \times 2 = 0.034$). So, for a two-tailed test, this p value (0.034) is compared with the α values 0.05 and 0.1. Hence, at $\alpha = 0.05$ and $\alpha = 0.1$, the null hypothesis will be rejected but at $\alpha = 0.01$, the null hypothesis will be accepted ($0.01 < 0.034$).

The p -value approach of hypothesis testing for large samples is sometimes referred to as the observed level of significance. The p -value defines the smallest value of α for which the null hypothesis can be rejected.

For Example 10.1, use the p -value method to test the hypothesis using $\alpha = 0.01$ as the level of significance. Assume that the sample mean is 10,200.

Example 10.2

Solution

In the light of this new sample mean $\bar{x} = 10,200$, the z value can be calculated as below:

$$z = \frac{10,200 - 10,000}{\frac{1200}{\sqrt{200}}} = \frac{200}{84.85} = 2.36$$

The observed test statistic is computed as 2.36. From the normal table, the corresponding probability area for z value 2.36 is 0.4909. So, the probability of obtaining a z value greater than or equal to 2.36 is $0.5000 - 0.4909 =$

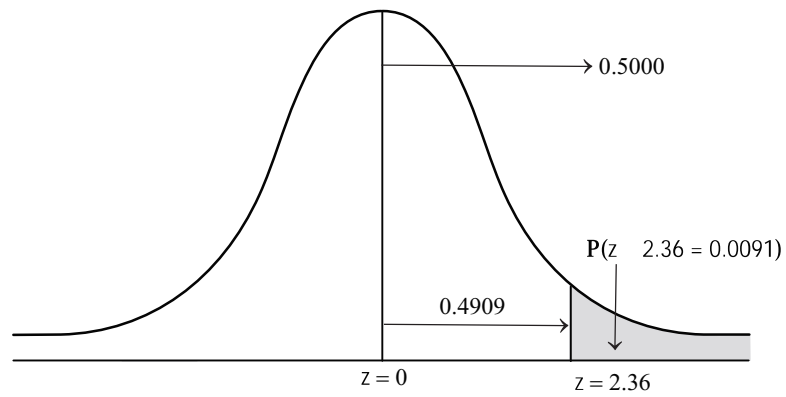


FIGURE 10.9
Probability of $p(z \geq 2.36)$

0.0091 (shown in Figure 10.9). For a two-tailed test, this value is multiplied by 2 (as discussed above). Thus, for a two-tailed test, this value is $(0.0091 \times 2 = 0.0182)$. So, the null hypothesis is accepted because $(0.01 < 0.0182)$. It has to be noted that for $\alpha = 0.05$ and $\alpha = 0.1$, the null hypothesis is rejected because $0.0182 < 0.05$ and $0.0182 < 0.1$.

10.6.2 Critical Value Approach for Hypothesis Testing

In Chapter 9, we have already discussed that z formula for estimating the population mean is given by

$$z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

Sample mean $\bar{x}$ can be greater than or less than the population mean. The above formula can also be written as

$$\mu = \bar{x} \pm z \frac{\sigma}{\sqrt{n}}$$

In the critical value approach for hypothesis testing, a critical $\bar{x}$ value, $\bar{x}_c$, and a critical z value, z_c , is determined and placed in the formula. After placing, the formula can be written as

$$\mu = \bar{x}_c \pm z_c \frac{\sigma}{\sqrt{n}}$$

After rearranging, this formula can be written as

$$\pm z_c = \frac{\bar{x}_c - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$\bar{x}_c = \mu \pm z_c \times \frac{\sigma}{\sqrt{n}}$$

This formula can be used for obtaining the lower and upper critical values. The critical value approach will be clearer with the help of Example 10.3.

Example 10.3

A cable TV network company wants to provide modern facilities to its consumers. The company has five-year old data which reveals that the average household income is Rs 120,000. Company officials believe that due to the fast development in the region, the average household income might have increased. The company takes a random sample of 40 households to verify this assumption. From the sample the average income of the households is calculated as 125,000. From historical data, population standard deviation is obtained as 1200. Use $\alpha = 0.05$ to verify the finding.

Solution

The null and alternative hypotheses can be set as below:

$$H_0: \mu = 120,000$$

$$H_1: \mu \neq 120,000$$

Since the sample size is 40, the z test must be used. The level of significance is taken as $\alpha = 0.05$. Using the critical value formula discussed earlier, we get:

$$\pm z_c = \frac{\bar{x}_c - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$$\pm 1.96 = \frac{\bar{x}_c - 120,000}{\frac{1200}{\sqrt{40}}} = \frac{\bar{x}_c - 120,000}{189.73}$$

$$\bar{x}_c = 120,000 \pm 371.87$$

This gives the lower and upper limits of the rejection region (as shown in Figure 10.10). Hence,

$$\text{Lower limit } \bar{x}_c = 120,000 - 371.87 = 119,628.13$$

$$\text{Upper limit } \bar{x}_c = 120,000 + 371.87 = 120,371.87$$

The result indicates that a sample mean of value greater than 120,371.87 and less than 119,628.13 will lead to the rejection of the null hypothesis. In this case, sample mean is calculated as 125,000 which leads to the rejection of the null hypothesis.

Note that for obtaining 95% confidence interval, $z_c \times \frac{\sigma}{\sqrt{n}} = 372$ (371.87 = 372 approximately) is deducted and added from the sample mean 125,000.

Hence, the lower limit is obtained as $125,000 - 372 = 124,628$ and upper limit is obtained as $125,000 + 372 = 125,372$ (as shown in Figure 10.11).

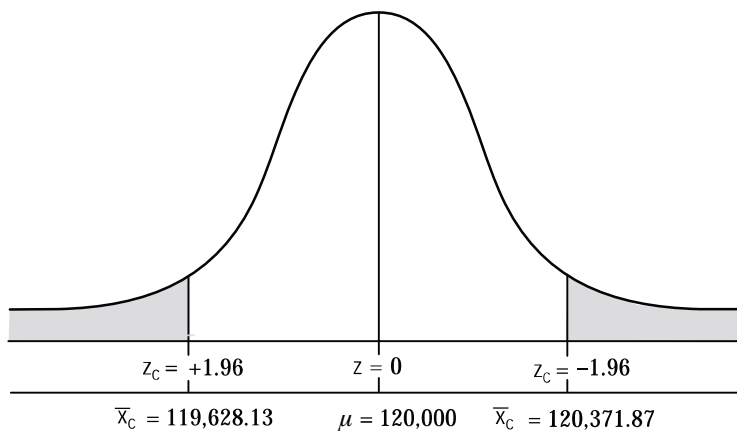


FIGURE 10.10
Critical value method for testing a hypothesis about the population mean for Example 10.3

One-Sample Z

Test of mu = 120000 vs not = 120000
The assumed standard deviation = 1200

N	Mean	SE Mean	95% CI	Z	P
40	125000	190	(124628, 125372)	26.35	0.000

FIGURE 10.11
Minitab output exhibiting 95% confidence interval for sample mean for Example 10.3